Homework 2

Collaboration with other students are encouraged; remember to name anyone you collaborate with and to write up your final solution by yourself.

When writing proofs, mathematical clarity is just as important as correctness and rigor. Please keep this in mind and try to adhere to proof writing conventions when typing up solutions.

1. Do you think loop invariants are unique? Why or why not? (This does not require a formal proof, nor are we checking for correctness. Sit with this for 10-15 minutes and write down your thoughts).

2. In your own words, explain why Fermat's little theorem holds. Consider a Carmichael-free universe.

$$a^{p-1} \equiv 1 \pmod{p}$$

Hint: Reference section 1.3 of the textbook

3. Construct generic merge sort in Dijkstra's notation, and analyze the runtime using $T(n)$.

4. If p is prime, how many elements of $\{0, 1, \ldots, p^n - 1\}$ have an inverse modulo p^n? Prove your answer.

5. In the last homework, we used modular exponentiation to assemble a polynomial time algorithm.

Algorithm: Modular Exponentiation

Modexp(X, Y, N)**:**
Input: n-bit integers N, X, Y, where Y is the exponent
Output: $X^Y \mod N$

$x, y, z := X, Y, 1$
do
$y \neq 0 \rightarrow$
if
y is even $\rightarrow x, y := x^2 \mod N,\ y/2$
y is odd $\rightarrow z, y := z \cdot x \mod N,\ y - 1$
return z

Find a satisfying loop invariant and prove its correctness.

6. Let N be a positive integer. We say that a is a *quadratic residue* modulo N if there exists x such that $a \equiv x^2 \bmod N$. Prove that if N is an odd prime and a is a nonzero quadratic residue modulo N, then there exist exactly two values x, y in $\mathbb{Z}_N$ such that $x^2 \equiv y^2 \equiv a \bmod N$.